Smart Parking System with Space Detection Using Deep Learning in a Mobile Application

Anis Amilah Shari
Faculty of Computer and Mathematical Science,
Universiti Teknologi MARA
Cawangan Melaka
Melaka, Malaysia
anisamilah@uitm.edu.my

Mohd Rahmat Mohd Noordin*
Faculty of Computer and Mathematical Science,
Universiti Teknologi MARA
Cawangan Melaka
Melaka, Malaysia
mrahmat.noordin@uitm.edu.my

Asma Shazwani Shari
Faculty of Business and Management
Universiti Teknologi MARA
Cawangan Kelantan
Kelantan, Malaysia
asma663@uitm.edu.my

Asiah Mat
Faculty of Computer and Mathematical Sciences
Universiti Teknologi MARA
Cawangan Kelantan
Kelantan, Malaysia
asiahmat@uitm.edu.my

Farah Muna Mohamad Ghazali
Faculty of Computer and Mathematical Sciences
Universiti Teknologi MARA
Cawangan Terengganu
Kampus Kuala Terengganu
Terengganu, Malaysia
farahmuna@uitm.edu.my

Siti Anisah Mohd Hatta
Academic of Language Studies
Universiti Teknologi MARA
Cawangan Kelantan
Kelantan, Malaysia
anisah780@uitm.edu.my

Norlida Razali
Academic of Language Studies
Universiti Teknologi MARA
Cawangan Kelantan
Kelantan, Malaysia
norlida8093@uitm.edu.my

Naufal Anas Shafie
Faculty of Computer and Mathematical Sciences
Universiti Teknologi MARA
Cawangan Melaka
Melaka, Malaysia
naudalanas03@gmail.com

2026 International Conference on Cognitive Systems and Computer Interaction (ICoSCI) | 979-8-3315-6813-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICOSCI66700.2026.11447969

***Abstract*—Parking congestion in university areas remains a persistent challenge, leading to extended search times, fuel waste, and frustration among students, staff, and visitors. Traditional campus parking systems lack real-time availability tracking and automated space allocation, resulting in these inefficiencies. This paper presents a smart parking system specifically designed for university campuses, integrating deep learning-based YOLOv8n for parking space detection and Dijkstra's algorithm for optimal route allocation. The system features a Flutter-based mobile application connected to a Flask backend, enabling users to view real-time parking availability, receive automatic assignments to the nearest available space, and navigate directly to their assigned spot. A custom dataset combining the PKLot dataset with campus- specific images was curated, preprocessed, annotated, and used for training and validation. The optimized YOLOv8n model achieved a mean Average Precision (mAP@0.5) of 0.994, precision of 0.998, and recall of 0.998. Experimental results demonstrate that the system significantly reduces parking search time, optimizes space utilization, and enhances user convenience. Future enhancements may include multi-zone coverage expansion, license plate recognition for security, and predictive analytics for peak-hour availability estimation.**

Keywords*—Smart parking system, YOLO, deep learning, university parking, automated space assignment, real-time detection.

I. Introduction

Parking inefficiencies have become a common problem in university environments due to the increasing number of vehicles and limited parking space availability. This imbalance leads to traffic congestion, long search times, unnecessary fuel consumption, and frustration among students, staff, and visitors. Several studies have highlighted that a significant portion of campus traffic is generated by drivers circulating in search of available parking spaces, particularly during peak hours [7], [10], [15]. These inefficiencies negatively impact campus mobility and contribute to environmental pollution. According to Shang et al. [1], parking demand during peak hours often exceeds available capacity, resulting in logistical and environmental challenges. Similar findings were reported in studies focusing on higher education institutions, where dissatisfaction among users was primarily due to the lack of real-time parking information and inefficient space allocation mechanisms [2], [10]. Traditional parking management systems typically rely on static permit-based allocation or manual monitoring, which are unable to adapt dynamically to fluctuating parking demand [3], [7]. Recent advancements in intelligent transportation systems and computer vision have enabled the development of smart parking systems capable of detecting parking occupancy in real time using camera-based solutions [11], [12], [16]. These inefficiencies negatively impact campus mobility and contribute to environmental pollution.

According to Shang et al. [1], parking demand during peak hours often exceeds available capacity, resulting in logistical and environmental challenges. Similar findings were reported in studies focusing on higher education institutions, where dissatisfaction among users was primarily due to the lack of real-time parking information and inefficient space allocation mechanisms [2], [10]. Traditional parking management systems typically rely on static permit-based allocation or manual monitoring, which are unable to adapt dynamically to fluctuating parking demand [3], [7]. Recent advancements in intelligent transportation systems and computer vision have enabled the development of smart parking systems capable of detecting parking occupancy in real time using camera-based solutions [11], [12], [16]. These systems reduce reliance on expensive physical sensors and offer scalable deployment, particularly suitable for university campuses. Furthermore, integrating mobile applications allows users to receive real-time availability updates and navigation assistance directly on their devices [15].

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on September 06,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.
979-8-3315-6813-9/26/$31.00 ©2026 IEEE

To address these limitations, this study proposes a smart parking system specifically designed for university campuses, integrating deep learning-based YOLOv8 for parking space detection and Dijkstra's algorithm for optimal route allocation. The system provides real-time availability visualization, automated parking space assignment, and efficient navigation through a mobile application, improving overall parking efficiency and user experience. Deep learning enables automatic feature extraction and robust pattern recognition, making it suitable for complex visual tasks such as parking space detection [9], [11], [12].

II. Approach

The proposed Smart Parking System was developed following the Waterfall software development model, selected for its structured and sequential design process. This model ensured each phase requirements analysis, system design, implementation, and testing was clearly defined and documented before progression to the next stage. The linear workflow-maintained consistency between planned system objectives and the final deployed prototype [6]. Smart parking systems represent a technology-driven solution aimed at enhancing parking management efficiency through real-time data acquisition, intelligent processing, and automated decision-making [11], [14].

By leveraging advancements in computer vision, Internet of Things (IoT), and cloud computing, these systems significantly reduce parking search time and traffic congestion while improving space utilization [7], [15]. Modern smart parking solutions are no longer limited to sensor-based approaches. Vision-based systems using deep learning models have gained popularity due to their cost-effectiveness, scalability, and ease of deployment [12]. Modern parking systems utilize sensor-based or vision-based detection techniques to provide real-time parking availability and guidance [7], [8], [14]. These systems utilize surveillance cameras to monitor parking areas continuously and apply object detection algorithms to determine parking space occupancy in real time. In university environments, smart parking systems are particularly beneficial due to fluctuating parking demand throughout the day. Studies show that campus-focused smart parking systems reduce average search time and improve user satisfaction when compared to traditional parking methods [10], [15].

A. System Overview

The system architecture consists of four major components:

- Camera input layer that captures live video of parking spaces.
- YOLOv8 detection model for real-time parking slot recognition.
- Flask backend server for processing detections and handling API communication.
- Flutter mobile application that displays current parking availability and provides navigation assistance using the Dijkstra algorithm.

The workflow begins with image acquisition from installed cameras. The YOLOv8 model detects each parking slot and classifies it as occupied or vacant. The Flask backend then sends this information to the mobile application through RESTful API endpoints [4]. Users can view the live status of available spaces, and the system automatically suggests the nearest free slot with a generated route.

B. System Design and Methodology

The design phase included the creation of use-case diagrams, system architecture diagrams, and a flowchart to visualize data movement between modules. Fig. 1 illustrates the main process.

- User authentication through the mobile interface.
- Camera capture and YOLOv8 detection for real- time space identification.
- Transmission of detection data to the backend for processing.
- Display of occupancy status on the user interface (green = vacant, red = occupied, blue = assigned).
- Path generation using the Dijkstra algorithm to guide the user to the nearest available slot.
- Confirmation of successful parking and database update.

This systematic flow ensures real-time updates, minimal delay between detection and display, and automatic synchronization between all modules.

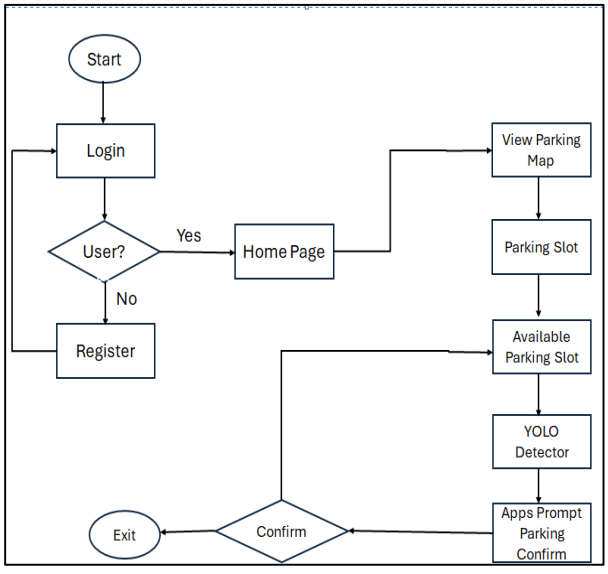


Fig. 1. System Flowchart

C. Dataset Development

A hybrid dataset was used, combining images from the PKLot dataset with photographs collected from UiTM Shah Alam campus parking areas. The campus images were captured under different lighting and weather conditions to improve generalization.

- Total images: approximately 2,500

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on September 06,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

- Classes: Occupied and Empty
- Annotation tool: Labeling for bounding box labeling in YOLO format.
- Preprocessing: resizing to 640×640 pixels, normalization, and augmentation (brightness, rotation, flipping).
- This dataset was divided into 70 % training, 20 % validation, and 10 % testing sets to ensure balanced performance evaluation.

D. Model Training and Optimization

The YOLOv8n model was trained using transfer learning on the prepared dataset. Training was conducted in Google Colab with GPU acceleration. Key hyperparameters:

- Epochs: 100
- Batch size: 16
- Learning rate: 0.001
- Input size: 640 × 640 pixels

The model demonstrated high detection performance with precision of 0.998, recall of 0.998, and mAP@0.5 of 0.994. The trained weights were exported and integrated into the Flask backend for inference during real-time operation.

E. Navigation and Backend Integration

The Dijkstra algorithm was implemented in the backend to compute the shortest path between the user's current position and the available parking slots. The algorithm used a weighted graph representation of the parking map, where each node corresponded to a path segment. The backend provides several RESTful API endpoints:

- /detect - process incoming frames through YOLOv8 and return slot status.
- /status - send real-time parking availability updates.
- /route - generate the optimal navigation path using Dijkstra's algorithm.

This backend continuously updates the mobile application, ensuring that parking data and route guidance remain synchronized.

III. DEVELOPMENT

The development phase centered on building and integrating the core modules of the smart parking system. The object detection capability was implemented using the YOLOv8n model, selected for its optimal balance of speed and accuracy, which is essential for real-time performance. To ensure high accuracy, the model underwent a two-stage training process. It was first pre-trained on the extensive public PKLot dataset before being fine-tuned on a custom dataset of 2,500 images specifically collected from the UiTM campus parking area.

This approach ensured the model was both robust and precisely adapted to the target environment. The finalized YOLOv8n model was then integrated into a central backend server built using the Flask framework. This server acts as the system's processing hub, where it continuously analyzes image frames from CCTV feeds to determine parking space occupancy. The detection results are used to maintain live status for every parking lot. For efficient data handling, the backend exposes several RESTful API endpoints, including /api/detect for processing images and /api/parking/status for providing real-time availability updates to the mobile application.

A critical feature of the system is its automated space assignment and navigation guidance, powered by Dijkstra's algorithm. When a user requests a parking space via the mobile app, the backend executes this algorithm on a pre-defined weighted graph that models the parking lot layout. The algorithm calculates the shortest possible path from a designated entrance to the nearest available slot, ensuring users are guided along the most efficient route. This path-finding logic is accessed by the mobile client through a dedicated /api/parking/assign endpoint.

The user interacts with the system through a cross-platform mobile application developed in Flutter. The application features an intuitive interface that displays an interactive, color-coded map of the parking facility. Parking slots are dynamically updated in real-time using green for available, red for occupied, and blue for the user's assigned slot. This live visual feedback is achieved by having the application listen for continuous updates from a cloud-based Firestore database, which is synchronized by the backend after every detection cycle.

Finally, all these components were woven into a seamless operational workflow. The system functions as a closed loop: cameras feed data to the backend, the YOLOv8 model performs detection, results update the database, and the mobile app reflects these changes instantly. Upon a user's request, the Dijkstra algorithm generates a navigation path, guiding the user to their assigned spot and concluding a fully automated parking process from discovery to arrival [13].

IV. EVALUATION

The proposed smart parking system was rigorously evaluated on two key fronts: the accuracy of the YOLOv8n object detection model and the functional performance of the integrated system. The model was trained and validated on a hybrid dataset (PKLot and UiTM), split into 70% for training, 20% for validation, and 10% for testing.

The fine-tuned YOLOv8n model demonstrated exceptional performance. It achieved a precision of 0.998 and a recall of 0.998, indicating a near-perfect ability to identify parking spaces correctly while minimizing false positives and negatives. The model's overall detection capability was confirmed by a mean Average Precision (mAP@0.5) of 0.994 and mAP@0.5:0.95 of 0.982, confirming high robustness even under stricter localization criteria.

The model's effectiveness is visually demonstrated in Figure 2, which shows its performance on real-world campus parking images. The model accurately draws bounding boxes and classifies spaces as space-occupied or space-empty with

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on September 06,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

high confidence, successfully handling challenges like varying vehicle sizes and partial occlusions.



Fig. 2. YOLOv8n Detection Results

Beyond quantitative metrics, the system was tested using live camera feeds. In this real-time setting, it maintained high classification accuracy across different lighting conditions and during peak occupancy. The Dijkstra-based navigation feature was functionally tested by users, who reported that the system consistently assigned the nearest available slot and generated optimal routes, significantly reducing parking search time compared to traditional methods [5].

These results collectively affirm that the system meets its core objectives. The YOLOv8n model provides highly precise and reliable real-time detection, while the integration with the Dijkstra algorithm ensures efficient space allocation, creating a streamlined and intelligent parking management solution.

V. Conclusion And Future Works

This paper presented a fully integrated smart parking system designed to alleviate congestion in university campuses. The system successfully leverages a fine-tuned YOLOv8n model for high-accuracy, real-time parking space detection and integrates the Dijkstra algorithm for optimal route allocation to the nearest available slot.

The implementation of a Flask backend with RESTful APIs and a Flutter-based mobile application creates a seamless user experience, allowing drivers to view live parking availability and receive guided navigation efficiently. Evaluation results demonstrate the system's robustness, with the model achieving a precision and recall of 0.998 and an mAP@0.5 of 0.994, confirming its reliability in real-world conditions.

For future work, the system's scalability will be enhanced to support multiple parking zones across a larger campus or urban area. Security and functionality can be augmented by integrating Automatic License Plate Recognition (ALPR) for access control and violation monitoring.

Furthermore, the user experience can be significantly improved by incorporating advanced features such as reservation-based booking, digital payment integration, and push notifications for session reminders. Exploring deployment with a distributed network of IP cameras could also provide more comprehensive coverage and system resilience. These improvements will collectively enhance the system's practicality, scalability, and overall impact on modern parking management.

Acknowledgment

The authors would like to express sincere appreciation and gratitude to Universiti Teknologi MARA (UiTM) Melaka for the support and assistance provided throughout the completion of this work. The corresponding author can be contacted via email at mrahmat.noordin@uitm.edu.my or asma663@uitm.edu.my.

References

[1] H. Shang, W. Liu, and H. Huang, “Parking demand analysis and evaluation at university campuses,” *Journal of Transportation Systems Engineering and Information Technology*, vol. 7, no. 2, 2007.

[2] F. I. Ibrahim, M. Fareez, and M. Aishah, “Parking spaces in Taylor’s University: Problems and solutions,” *Environment- Behaviour Proceedings Journal*, vol. 5, no. 13, pp. 323–331, 2020. https://doi.org/10.21834/e-bpj.v5i13.2035

[3] N. Nadimi, S. Afsharipoor, and A. M. Amiri, “Parking Demand vs Supply: An Optimization-Based Approach at a University Campus,” *Journal of Advanced Transportation*, vol. 2021. https://doi.org/10.1155/2021/7457021

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only LookOnce: Unified, Real-Time Object Detection,” in *IEEE CVPR*, 2016,pp. 779–788.

[5] H. Cao, X. Wang, and F. Wang, “Application Case Study Based on Dijkstra Algorithm,” in *IEEE ICIPCA 2024*, pp. 1149–1154. https://doi.org/10.1109/ICIPCA61593.2024.10709108

[6] J. J. Barriga et al., “Smart Parking: A Literature Review from the Technological Perspective,” *Applied Sciences*, vol. 9, no. 21, 2019. https://doi.org/10.3390/app9214569

[7] A. Amato, M. Carrara, C. Cimino, A. Nanni, and G. Vaglini, “Car parking occupancy detection using smart camera networks and deep learning,” IEEE Symposium on Computers and Communications (ISCC), 2016, pp. 1212–1217. https://doi.org/10.1109/ISCC.2016.7543903

[8] S. Suhr and J. Jung, “Sensor fusion-based vacant parking space detection and tracking,” IEEE Transactions on Intelligent Transportation Systems, vol. 15, no. 1, pp. 21–36, 2014. https://doi.org/10.1109/TITS.2013.2272090

[9] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” IEEE International Conference on Computer Vision (ICCV), 2017, pp. 2980–2988. https://doi.org/10.1109/ICCV.2017.324

[10] G. T. Salleh, M. A. Rahman, and N. A. Zainal, “Smart campus parking system using IoT and mobile application,” International Journal of Advanced Computer Science and Applications, vol. 10, no. 6, pp. 257–264, 2019. https://doi.org/10.14569/IJACSA.2019.0100634

[11] U. Asghar, M. S. Khan, and S. H. Shah, “Real-time vehicle detection for smart parking systems using deep learning,” Journal of Ambient Intelligence and Humanized Computing, vol. 12, pp. 1–14, 2021. https://doi.org/10.1007/s12652-020-02705-8

[12] A. Rehman, T. Saba, and A. Al-Shahri, “Deep learning-based parking occupancy detection using CCTV images,” Computers, Materials & Continua, vol. 65, no. 1, pp. 73–89, 2020. https://doi.org/10.32604/cmc.2020.010456

[13] R. Dijkstra, “A note on two problems in connexion with graphs,” Numerische Mathematik, vol. 1, pp. 269–271, 1959. https://doi.org/10.1007/BF01386390

[14] M. K. Hasan, A. Islam, and M. M. Rahman, “IoT-based smart parking management system using image processing,” International Conference on Electrical, Computer and Communication Engineering (ECCE), 2019, pp. 1–6. dijkstrahttps://doi.org/10.1109/ECACE.2019.8679252

[15] J. Liu, Y. Li, and L. Zhang, “Mobile-based smart parking system with real-time data analytics,” Future Generation Computer Systems, vol. 94, pp. 43–56, 2019. https://doi.org/10.1016/j.future.2018.11.013

[16] A. Bochkovskiy, C. Wang, and H. M. Liao, “YOLOv4: Optimal speed and accuracy of object detection,” arXiv preprint arXiv:2004.10934, 2020.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on September 06,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.